

```

pins high
INTCON.T0IE = 1 'enable Timer0
interrupts
INTCON.GIE = True 'enable all
un-masked interrupts
OPTION_REG.T0CS = False 'set Timer0
clock source to internal instruction
cycle clock
End
On Interrupt 'interrupt routine
PORTB = PORTB - 1 'decrement the
value on PORTB
INTCON.T0IF = 0 'enable new TMR0
interrupts
Resume

```

Go to File > Load Program, or press Ctrl+L. Browse and select the file timer0.hex from the window, and open it. The path of the file now appears in the box next to Program Location, and the file will be loaded into IC program memory.

Select the PIC model from Options > Select Microcontroller (or simply click on the box next to Microcontroller in main window) and select the appropriate microcontroller (PIC16F84 in this example) from the window.

To open the graphical view of microcontroller pinout and logic states, click on Tools > Microcontroller View. Select the simulation rate to Extremely Fast from Rate in menu bar (or press Ctrl+F5.) The simulation will start immediately when you select Simulation > Start, or press F1. To stop the simulation any time, click on Simulation > Stop (or press F3).

It is to be noted that, in order to compile this file for other PIC microcontroller models, you may have to change T0IE and T0IF bit names in the source code to TMR0IE and TMR0IF. The datasheet of the model you want to use would have the correct bit names for the INTCON register.

Developer	Vladimir Soso, Oshon Software Solutions (www.oshonsoft.com)
Operating system	Windows Reported to be working with Linux under Wine project
Latest version	6.96
File size	2.13MB
Licence type	Shareware/Trialware/Demoware

Example 2: 7-segment LED display simulation

The following Basic program makes the numbers 0 to 100 display on a 7-segment LED display:

```

Dim digit As Byte 'input variable for
GETMASK subroutine
Dim digit1 As Byte 'current high
digit
Dim digit2 As Byte 'current low digit
Dim mask As Byte 'output variable
from GETMASK subroutine
Dim mask1 As Byte 'current high digit
mask
Dim mask2 As Byte 'current low digit
mask
Dim i As Byte
Dim phase As Bit
Symbol dlenable = PORTC.0 'enable
line for higher 7-segment display
Symbol d2enable = PORTC.1 'enable
line for lower 7-segment display
TRISB = %00000000 'set PORTB pins as
outputs
TRISC.0 = 0 'set RC0 pin as output
TRISC.1 = 0 'set RC1 pin as output
dlenable = False
d2enable = False
mask1 = 0
mask2 = 0
phase = 0
INTCON.T0IE = 1 'enable Timer0
interrupts
INTCON.GIE = 1 'enable all un-masked
interrupts

```

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16F76	16F689	16F884	16F616
16F77	16F690	16F886	16F707
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16F84A	16F747	16F913	16F720
16F87	16F767	16F914	16F721
16F88	16F777	16F916	16F722
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16F630	16F872		16F726
16F631	16F873		16F727
16F636	16F873A		16F753
16F639	16F874		16F785

```

OPTION_REG.T0CS = 0 'set Timer0 clock
source to internal instruction
cycle clock
loop:
For i = 0 To 99
digit1 = i / 10 'get current high
digit
digit2 = i Mod 10 'get current low
digit
TMR0 = 0 'reset Timer0 to prevent its
interrupt before both masks are
determined
digit = digit1

```



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